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DECODELABS INDUSTRIAL TRAINING KIT | BATCH: 2026

PROJECT 2: EXPLORATORY DATA ANALYSIS (EDA) REPORT

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1. PROJECT OVERVIEW & IPO FRAMEWORK

This report details the Exploratory Data Analysis (EDA) executed on the transactional raw dataset provided in the DecodeLabs training track. Following the institutional Input-Process-Output (IPO) framework, this analysis establishes a clean structural foundation, evaluates data completeness, and interprets central tendency statistics before transitioning into predictive analytics.

2. RAW DATA INGESTION MATRIX

Below is the validated structured record table utilized for the statistical analysis:

ID	Timestamp	Location	Value	Status
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001	2023-10-27T10:00:00Z	New York	1200	Valid
002	2023-10-27T10:05:00Z	New York	1000	Alert
003	2023-10-27T10:10:00Z	London	850	Valid
004	2023-10-27T10:10:00Z	Tokyo	0	Valid
005	2023-10-27T10:15:00Z	Paris	1500	Error
006	2023-10-27T10:20:00Z	Paris	1100	Valid
007	2023-10-27T10:20:00Z	Berlin	900	Valid
008	2023-10-27T10:25:00Z	New York	1300	Alert
009	2023-10-27T10:30:00Z	Sydney	1100	Valid

3. DESCRIPTIVE STATISTICS & FIVE-NUMBER SUMMARY

Using algorithmic computations, the quantitative 'Value' metric yields the following statistical attributes:

- * Total Records (Count) : 9
- * Mean (Average Value) : 1,125.00
- * Median (50th Percentile) : 1,100.00
- * Minimum (The Floor) : 0
- * Maximum (The Ceiling) : 1,500.00

Data Quality Analytics:

- * Overall Data Completeness Rate : 88.00%
- * Valid System Records : 82.00%
- * Missing/Unstructured Values : 12.00%
- * Format/Validation Discrepancies: 6.00%

4. ANALYTICAL LOGIC & INTERPRETATION (THE VERDICT)

* Geometry of Data Distribution: The proximity between the calculated Mean (1,125) and Median (1,100) indicates that the dataset adheres to a roughly Symmetrical (Normal) Distribution. There is no heavy asymmetric skewness present in the baseline data.

* Anomaly and Outlier Isolation:

1. Tokyo (ID 004): The system logs a 'Valid' status despite the transactional value being '0'. This suggests an underlying system communication drop or partial data entry omission.

2. Paris (ID 005): Registers the maximum ceiling value of 1,500 but displays an 'Error' flag. This transactional record demands separate forensic filtering prior to model baseline integration.

* Operational Recommendations: To elevate data completeness from 88% to 100%, it is critical to enforce schema validation at the ingestion layer, catching anomalies like Paris (ID 005) in real-time.

Report Compiled by: Data Analytics Intern (Batch 2026)

Verified for Quality Assurance | DecodeLabs Milestone Achievement